

A Reimplementation of DreamBooth and Extensions: LoRA & Text-to-Video

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GitHub: <https://github.com/EVAGISH/cs4782-final-project>

Introduction

DreamBooth proposes a method to “personalize” text-to-image diffusion models using only 3-5 images of a specific subject. The approach fine-tunes a pretrained model to bind a unique rare-token identifier to the subject, allowing it to synthesize novel photorealistic images of that subject in diverse contexts guided by text prompts. The paper’s key contribution is a class-specific prior preservation loss, which prevents language drift and maintains output diversity during fine-tuning. We re-implemented Dreambooth with Stable Diffusion and attempted some interesting extensions, including finetuning with LoRA and applying the method to text-to-video models. Lastly, we experimented with classifier guidance to achieve personalization without weight updates, showing that naive implementations of such techniques fail.

Chosen Result

We evaluate against three metrics introduced by Ruiz et. al.

DINO measures instance-level subject identity; because

DINO is a self-supervised vision transformer trained without

class labels, its features discriminate between different

members of the same class, making it the primary fidelity

signal for subject-driven generation. **CLIP-I** is a coarser,

class-level semantic similarity score and serves more as a

sanity check that the generated image is of the right class of

object. **CLIP-T** is the text-image cosine similarity with the

prompt and measures prompt fidelity; it acts as a counter-metric to DINO and detects the failure mode where

over-aggressive fine-tuning can result in producing a high-fidelity image of a subject that ignores the requested contexts.

Method	DINO ↑	CLIP-I ↑	CLIP-T ↑
Real Images	0.774	0.885	N/A
DreamBooth (Imagen)	0.696	0.812	0.306
DreamBooth (Stable Diffusion)	0.668	0.803	0.305
Textual Inversion (Stable Diffusion)	0.569	0.780	0.255

Table 1. Subject fidelity (DINO, CLIP-I) and prompt fidelity (CLIP-T, CLIP-T-L) quantitative metric comparison.

Methodology

For text-to-image tasks, we primarily conducted our experiments with Stable Diffusion v1.5, a pre-trained TTI latent diffusion model. Note that Imagen, used in the original paper, is closed source. We kept the text encoder and the VAE of Stable Diffusion frozen, fine-tuning only the UNet (~859.9M parameters). Throughout our experiments, we used the same rare token from the original paper - ‘sks’ - from the CLIP tokenizer. We conducted experiments for four subjects: backpack, bear plushie, cat, and dog, each with 5 subject training images. For each class, we generated 200 class prior images for the prior preservation loss. Training used the exact DreamBooth objective, a linear combination of two standard diffusion denoising objectives, one for a subject image and one for a class prior image. We used a lambda of 1.0 for the PPL weighting, just like in the original paper. We conducted all of our training with fp32 precision, running full-finetuning jobs for approximately 1000 steps with a learning rate of 5e-6.

Methodology: Extensions

Our first extension was to use LoRA throughout the DreamBooth finetuning process. We reasoned that LoRA would have a similar regularizing effect as the prior preservation loss and this proved to be correct as LoRA performed better with a lower PPL weighting (0.5) than what was optimal for full fine-tuning (1.0). We only performed LoRA on the Q, K, V projections in the 32 attention blocks of Stable Diffusion, yielding 96 total A/B pairs for LoRA. We used rank 16 for the LoRA pairs after some experimentation with different choices of rank, discovering that lower choices of rank led to degradation of subject fidelity. We also noticed that training with LoRA required a larger learning rate, and using the smaller learning rate that was set for full fine-tuning resulted in little progress towards the training objective.

Our next extension was to adapt the DreamBooth formulation to ModelScopeT2V, a text-to-video model very similar to Stable Diffusion; we were primarily interested in determining if DreamBooth would scale to much higher dimensional latent spaces. ModelScopeT2V represents entire videos as blocks of image frame latents and the architecture is a spatio-temporal UNet that can adapt to varying numbers of input frames. The DreamBooth formulation from Stable

Diffusion goes through easily, with denoising objectives just computed on the higher-dimensional block latents. We were able to achieve robust subject binding without catastrophic forgetting using DreamBooth on the video domain.

Lastly, we ran some experiments to determine whether weight-level updates were necessary to achieve the personalization of DreamBooth. The most naive way to do so is using classifier guidance. For our guidance term, we added an auxiliary L2 loss between the current noisy latent and a target latent, forward-diffused to the same noise level via the scheduler. We tried four target-selection strategies: 1) the pooled mean of all subject-image latents, 2) a randomly chosen single dataset latent fixed per sample, 3) a freshly resampled dataset latent at every denoising step, and 4) the closest noised subject image latent to the current latent in the denoising process. Strategies 1, 3, and 4 resulted in blurriness and strategy 2 resulted in mode collapse. We experimented with different auxiliary loss weightings and schedules per strategy, but all attempts ultimately fell into the same two failure modes. Thus, classifier guidance largely failed, supporting the hypothesis that weight-level updates like in DreamBooth are necessary to achieve personalization with Stable Diffusion.

Results & Analysis

Our implementation of the DreamBooth architecture achieves scores on DINO, CLIP-I, and CLIP-T that are broadly comparable to those reported in the original paper. Our Full FT matches the paper on subject semantics (CLIP-I 0.77 vs. 0.80) and actually exceeds it on prompt fidelity (CLIP-T 0.33 vs. 0.30). Our DINO score sits 0.07 below the paper's 30-subject average, which falls within their reported per-subject variance and is consistent with the non-uniform identity recovery we observed across different subjects and classes.

Our LoRA experiments demonstrate that DreamBooth's performance can be achieved with parameter efficient fine-tuning. By training only 0.28% of the UNet backbone's parameters - a 357 times reduction - we recover approximately 86% of Full FT's subject fidelity score on DINO while matching its CLIP-T. Interestingly, prompt fidelity scores actually improve slightly under LoRA, suggesting that the low-rank update naturally exhibits some of the regularizing effects that the prior preservation loss is designed to provide. The resulting checkpoints are also 445 times smaller (4.3 GB to 9.6 MB), making them substantially easier to store and share. Wall-clock and VRAM savings were more modest at 25% and 36% respectively.

Finally, our prior-loss weight ablation on the dog subject revealed that LoRA prefers a weaker prior than Full DreamBooth, with both subject metrics peaking at $\lambda=0.5$. At $\lambda=1.0$ the model over-regularizes and under-fits the subject, while $\lambda=0.1$ begins to drift on prompts - validating the paper's claim that prior preservation regularizes against language drift, while showing that the optimal weight shifts when using a low-rank update.

Reflections

One of our central takeaways was the tradeoff between subject identity and prompt fidelity, which the original authors' choice of DINO and CLIP-T metrics elucidates well. We also observed that identity recovery is highly non-uniform across subjects and classes - a single aggregate DINO score tends to obscure this, but reviewing individual subjects' generated images makes it clear.

We were also impressed by LoRA's impact on the number of trained parameters, though its effect on wall-clock training time was more nuanced. LoRA recovered most of full fine-tuning's capability at a small fraction of the parameters, but the reduction in training time was more modest than expected: the frozen parts of the UNet still run on every step, so per-step compute does not decrease in proportion to the parameter savings.

Future work could proceed along several complementary directions. A natural next step is training on multiple subjects simultaneously, which would involve identifying additional rare tokens and binding them to distinct subjects during fine-tuning. Furthermore, to build on our extensions, future work may also include alternative implementations of the prior preservation loss, further hyperparameter tuning for LoRA, implementing DreamBooth for rectified flow transformers, and extending our video generation experiments to newer base models with different latent representations.

References

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